

Amazon E-Commerce Sales Analytics

Data Analytics & Business Analytics Project Report

Analysis Period

01 January 2011 – 31 December 2014

Scope

51,290 order-line records across four fiscal years, spanning 147 countries and multiple product categories

\$12.64M

Total Sales

\$1.47M

Total Profit

11.61%

Profit Margin

178.3K

Units Sold

25,035

Total Orders

1,590

Customers

Executive Summary

This report presents a comprehensive analysis of Amazon e-commerce sales data from 2011 to 2014, covering 51,290 order-line records. The findings reveal strong revenue growth, stable profitability, and clear opportunities for margin improvement across categories and geographies.

Financial Highlights

- Total Sales: **\$12.64M** | Total Profit: **\$1.47M**
- Profit Margin: **11.61%** — stable throughout the period
- Sales grew **90.3%** from \$2.26M (2011) to \$4.30M (2014)
- Units Sold: 178.3K across 25,035 orders from 1,590 customers

Category & Market Leaders

- **Technology** leads by revenue at \$4.74M with 13.99% margin
- **APAC** is the strongest market at \$3.59M in sales
- **EMEA** is the weakest performer at only 5.45% margin
- **Canada** leads all markets with a 26% profit margin

Revenue Growth

90.3% increase over four years — from \$2.26M to \$4.30M

Margin Stability

Profit margin held at ~11.6% across all four years despite rapid growth

Key Risk

Tables sub-category generates a -\$64.1K loss; EMEA margin at 5.45%

Project Objectives & Methodology

This project was designed to deliver a structured, data-driven view of Amazon's e-commerce performance from 2011 to 2014, enabling actionable business decisions through rigorous analysis and interactive visualization.

1

Measure

Overall sales, profit, quantity, orders, customers, and shipping performance across the full four-year period

2

Identify

High and low-performing categories, sub-categories, markets, countries, and states by revenue and margin

3

Analyze

Yearly growth and profitability trends from 2011 through 2014 to surface structural patterns

4

Evaluate

Customer segments and shipping modes to inform operational and commercial planning

5

Enable

Interactive Power BI analysis with category-level drill-down across all dimensions

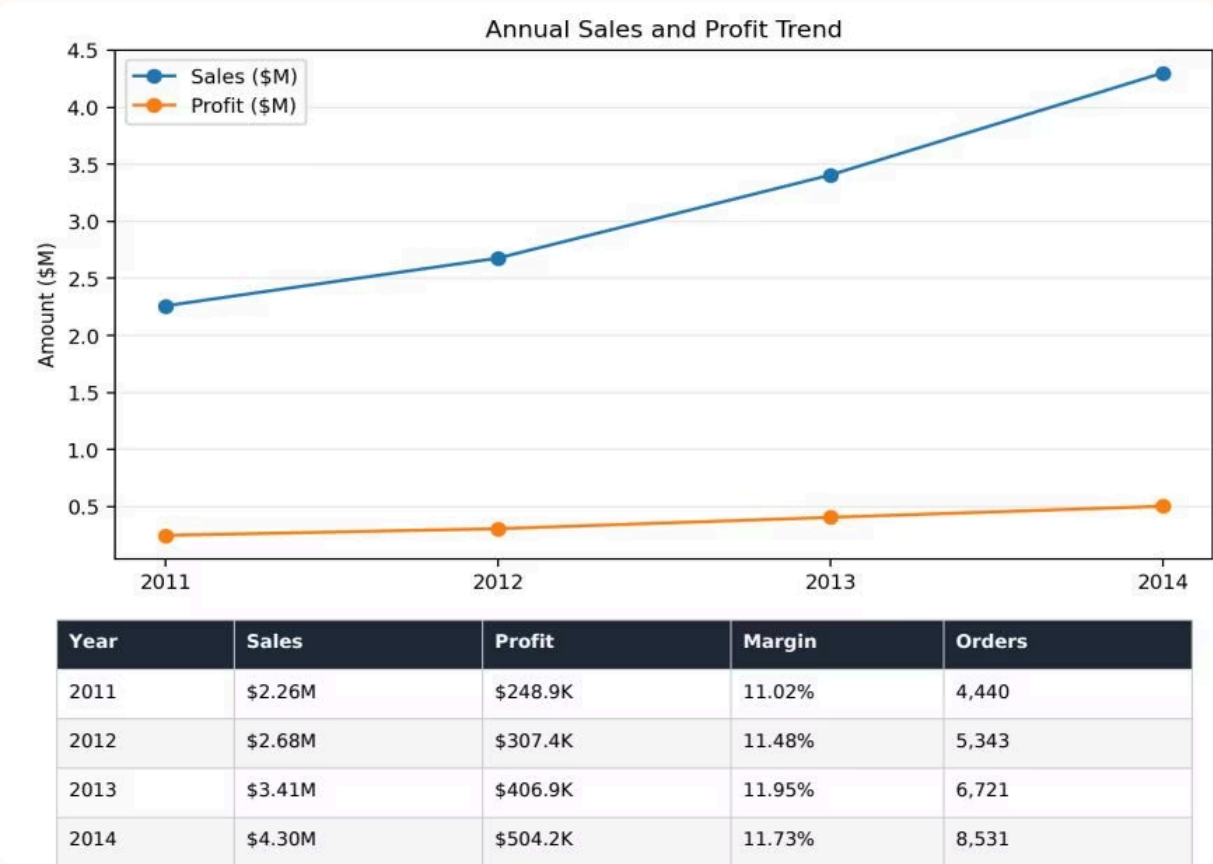
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Translate

All findings into practical, prioritized business recommendations for leadership

Sales & Profitability Trends (2011–2014)

Revenue grew consistently year-over-year, nearly doubling over the four-year period. Profit margins remained remarkably stable at approximately 11.6%, indicating disciplined cost management even as the business scaled rapidly.



Sales (\$K) Profit (\$K)



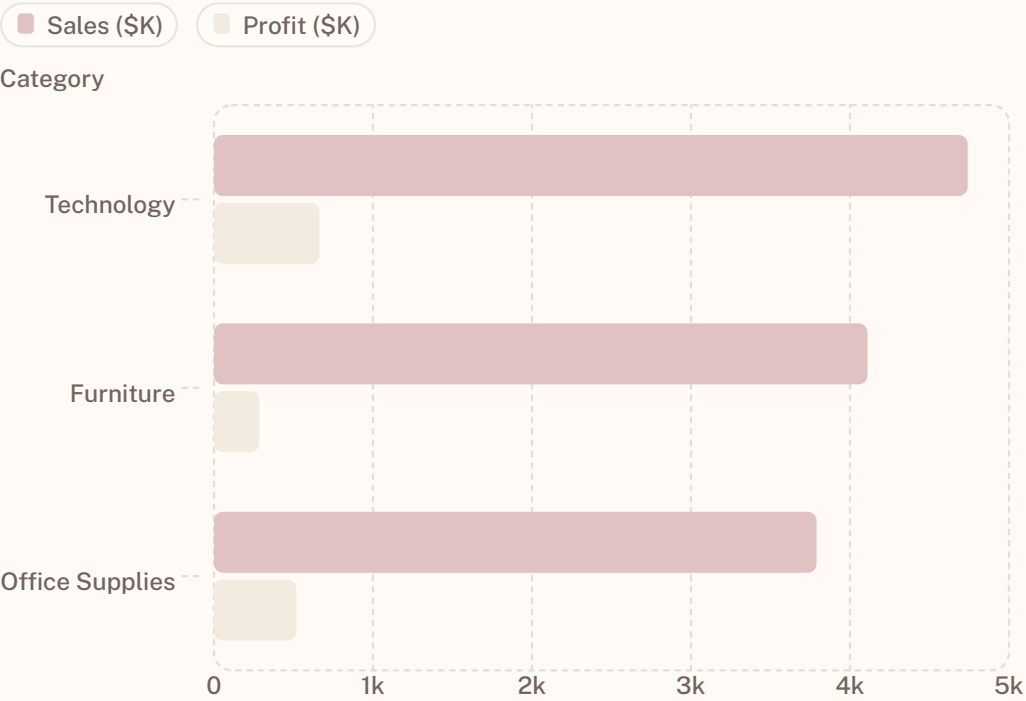
Year-by-Year Breakdown

Year	Sales	Profit	Margin
2011	\$2.26M	\$248.9K	11.02%
2012	\$2.68M	\$307.4K	11.48%
2013	\$3.41M	\$406.9K	11.95%
2014	\$4.30M	\$504.2K	11.73%

Profit margin held within a narrow 11.02%–11.95% band across all four years — a sign of consistent operational discipline.

Category Performance Analysis

Three product categories drive the entire business. Technology leads in both revenue and margin, while Furniture — despite generating \$4.11M in sales — delivers the weakest profitability at just 6.94%, signaling structural cost and pricing challenges.



Technology — 13.99% Margin

\$4.74M sales | \$663.8K profit. Highest margin category. Priority for continued investment and expansion.

Office Supplies — 13.69% Margin

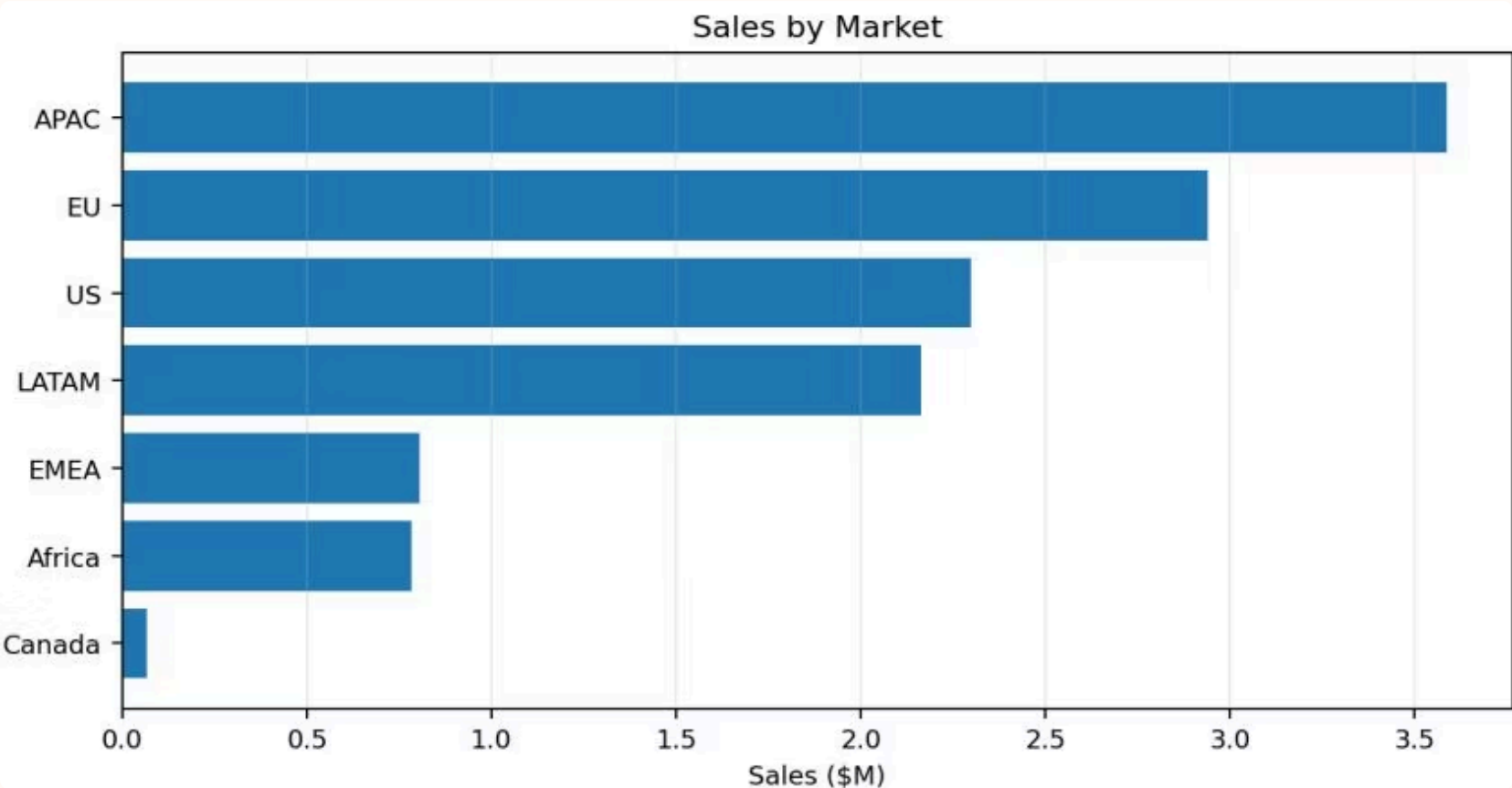
\$3.79M sales | \$518.5K profit. Strong profitability despite lower revenue volume than Furniture.

Furniture — 6.94% Margin ⚠️

\$4.11M sales | \$285.2K profit. Highest revenue but lowest margin. Requires urgent pricing and cost review.

Geographic & Market Performance

APAC dominates by revenue while Canada leads all markets in profitability. EMEA represents a significant margin concern — generating \$806.2K in sales but only a 5.45% margin, the lowest of any region.



Market Sales & Margins

Market	Sales	Margin
APAC	\$3.59M	—
EU	\$2.94M	12.69%
US	\$2.30M	12.47%
LATAM	\$2.16M	—
EMEA	\$806.2K	5.45% ⚠️
Canada	—	26.00%

Top Countries & States

Top Countries by Sales

United States (\$2.30M) · Australia (\$925.2K) · France (\$858.9K)

Top States by Sales

England (\$485.2K) · California (\$457.7K) · Ile-de-France (\$317.8K)

Margin Leaders

Canada (26%) leads all markets — followed by EU (12.69%) and US (12.47%)

Customer Segments & Shipping Modes

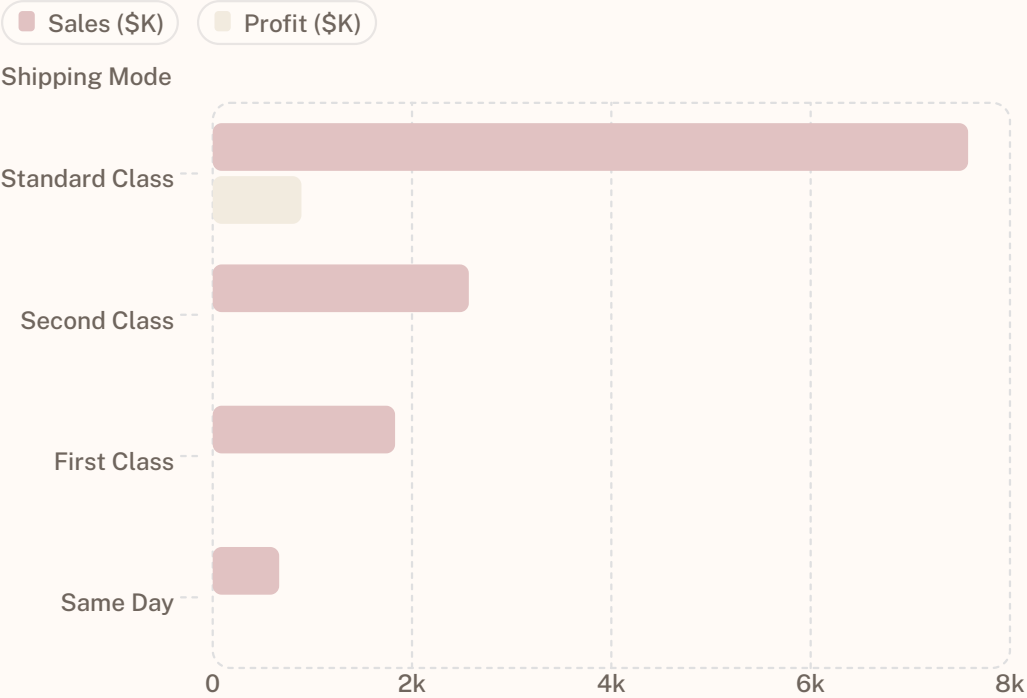
All three customer segments deliver nearly identical profit margins (~11.5–12%), indicating consistent pricing discipline across the customer base. Shipping mode analysis reveals Standard Class as the dominant channel — making it the highest-leverage area for operational efficiency gains.

Customer Segment Performance

Segment	Sales	Margin
Consumer	\$6.51M	11.51%
Corporate	\$3.82M	11.54%
Home Office	\$2.31M	11.99%

Home Office achieves the highest margin (11.99%) despite the smallest revenue base — suggesting a favorable product mix or lower discount rates in this segment.

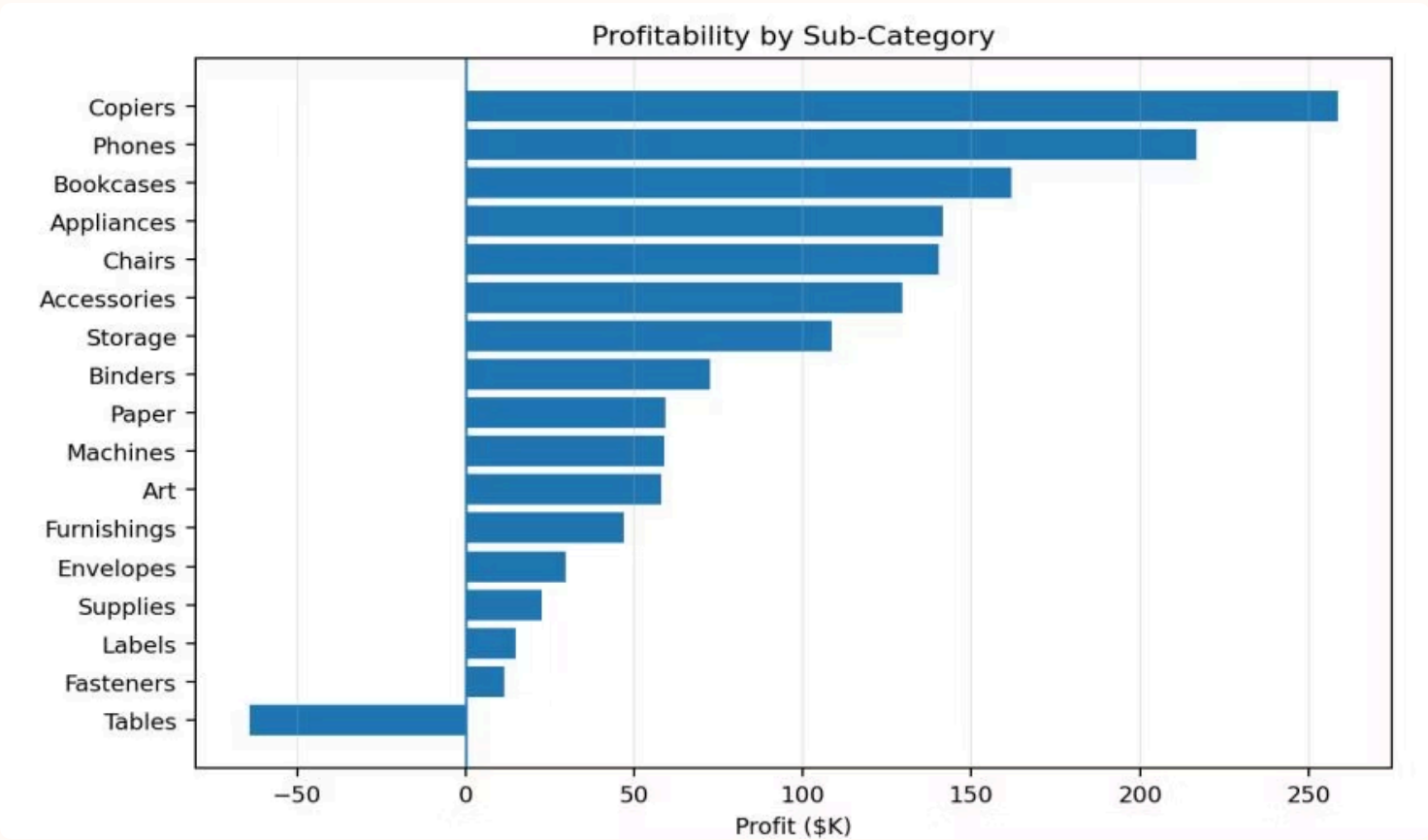
Shipping Mode Breakdown



Standard Class accounts for \$7.58M in sales and \$890.6K in profit — efficiency improvements here will have the largest financial impact of any shipping initiative.

Sub-Category Profitability & Risk

Sub-category analysis reveals a wide dispersion in profitability — from high-margin consumables like Labels (20.45%) to the critically loss-making Tables sub-category at -\$64.1K. Addressing these outliers is essential to improving overall portfolio health.



Top Profit Performers

Copiers

\$260K profit

Phones

\$215K profit

Bookcases

\$160K profit

Risk & Margin Concerns

Tables — CRITICAL LOSS: -\$64.1K

\$757K in sales yet generates a net loss. Pricing, discounting, and supplier costs must be reviewed immediately.

Machines — 7.56% Margin

Below-average margin signals potential over-discounting or high cost of goods.

Supplies — 9.29% Margin

Underperforming relative to category peers. Pricing strategy review recommended.

Furnishings — 12.18% Margin

Acceptable but below Technology and Office Supplies benchmarks.

High-Margin Winners

Sub-Category	Margin
Labels	20.45%
Envelopes	17.32%
Art	15.58%

Key Business Problems Identified

The analysis surfaces six distinct business problems requiring management attention. These range from structural margin issues in Furniture to operational risks from discounting and returns — each with a measurable financial impact.

1 Furniture Margin Gap

Furniture's 6.94% margin significantly lags Technology (13.99%) and Office Supplies (13.69%). The gap represents millions in unrealized profit on \$4.11M in annual revenue.

2 Tables Sub-Category Loss

Tables generates a **-\$64.1K loss** despite \$757K in sales — a critical profitability drain that must be addressed through pricing restructuring or product discontinuation.

3 EMEA Margin Underperformance

EMEA's 5.45% margin is the lowest across all regions. With \$806.2K in sales, even modest margin improvement would yield significant profit uplift.

4 Return Rate Opportunity

A return rate of **4.69%** creates a measurable opportunity for quality, logistics, and customer experience improvement across categories and markets.

5 Discount Monitoring Risk

A weighted average discount of 0.11% requires active monitoring. Uncontrolled discounting — particularly in Furniture — is a likely driver of negative-profit transaction lines.

6 Geographic Profit Gaps

Certain geographic areas generate sales volume but negative profitability — indicating that revenue growth alone is not a sufficient performance metric at the regional level.

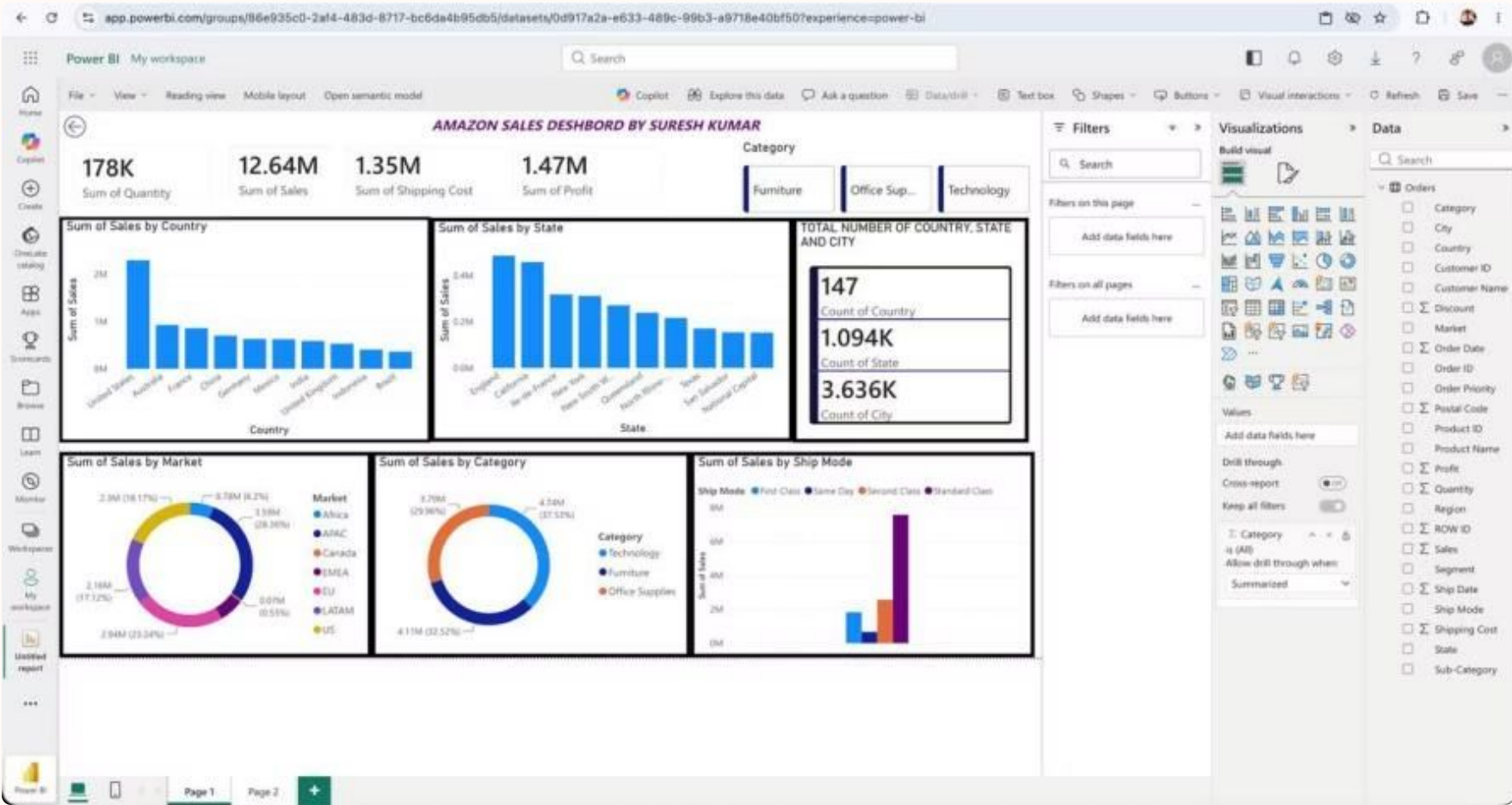
Strategic Recommendations

Seven prioritized recommendations address the core business problems identified in the analysis. Each recommendation is directly tied to a specific data finding and is designed to improve margin, reduce risk, or unlock growth.

1	Optimize Furniture Pricing Review pricing, supplier costs, freight, and discounts — with immediate focus on the Tables sub-category to eliminate the -\$64.1K loss.
2	Control Discounting Introduce margin-based discount thresholds. Approve discounts only when contribution margin remains positive. Flag exceptions for management review.
3	Prioritize High-Margin Growth Expand Technology and Office Supplies investment where both revenue and profitability are strong. Reallocate resources from low-margin Furniture lines.
4	Improve EMEA Economics Analyze freight costs, product mix, and discount patterns in EMEA to close the margin gap from 5.45% toward the 12%+ achieved in EU and US markets.
5	Investigate Returns Analyze returned products by category, market, and segment to identify quality issues and reduce the 4.69% return rate.
6	Regional Pricing Strategy Use market-level profitability and logistics cost data to guide localized pricing decisions — particularly in EMEA and LATAM where margins are under pressure.
7	Build Management Alerts Add Power BI alerts for negative-profit products, declining margins, and unusual discount patterns to enable proactive management intervention.

Interactive Power BI Dashboard

All findings from this report are surfaced through an interactive Power BI dashboard, enabling dynamic exploration of the data across categories, markets, and time periods. The dashboard covers **147 countries**, **1,094 states**, and **3,636 cities**.



Dashboard KPI Cards



Dashboard Capabilities

- Geographic performance charts show sales by Country and State
- Market and Category contribution visualized through donut charts
- Category slicer enables dynamic comparison of Furniture, Office Supplies, and Technology
- All metrics and charts update automatically when a category filter is applied
- Full geographic coverage: 147 countries, 1,094 states, 3,636 cities